

Technical Master Compliance & Safety Report: PDM (Power Distribution Module)

Parameter	Specification
Vehicle Subsystem	Power Distribution Module (PDM - Smart Solid-State LV Power Distribution)
Production Target	PDM_FW (ESP-IDF v5 + FreeRTOS SMP)
Applied Standards	MISRA-C:2012, ISO 26262 (ASIL-B Principles), Zero Dynamic Memory, 100 Hz Deterministic FreeRTOS
CI/CD Build & Test Status	PASSED (Compilation with -Werror , static analysis, and 100% host unit tests passing)

1. Executive Summary & System Architecture

The **Power Distribution Module (PDM)** serves as the intelligent solid-state electronic fuse box and power management hub for all Low-Voltage (LV) vehicle subsystems:

- Independent 12-Channel Solid-State MOSFET Power Switching:** Controls low-side/high-side power stages powering critical subsystems (Motor Control Unit, Inverter Logic, Cooling Pumps, Steering Wheel Electronics, Telemetry, Inertial Navigation SBG, ECU, etc.).
- Multi-Channel Precision Current Sensing:** Multiplexed acquisition across 12 resistive shunts ($R_{shunt} = 0.05\ \Omega$, operational amplifier gain $G = 20\ V/V$) via the **CD74HC4067** analog multiplexer, plus 2 dedicated high-current Hall-effect sensors (Shutdown Circuit 10A and Cooling Fans 30A).
- Ultra-Fast Electronic Overcurrent Protection:** Instantaneous trip if measured current exceeds +130% of rated threshold, combined with a deterministic 3-sample inrush debouncing filter on inductive channels (Inverter and Steering Wheel).
- Low-Voltage Battery Undervoltage Protection:** Automatic global load shutdown if $V_{bat} < 5.0V$ persists for $> 200\ ms$, safeguarding battery chemistry against destructive deep discharge.
- Deterministic CAN / TWAI Telemetry (500 kbps):** Emits complete switching state and milliampere current readings at 10 Hz across IDs 0x001 through 0x006, alongside dedicated diagnostic trouble codes on ID 0x501.
- Safety OTA Interlocking:** Rejects remote firmware updates whenever vehicle is in active READY_TO_DRIVE mode.

2. Exhaustive Fault Management Matrix (DTC & Safe States)

The PDM firmware features a safety-critical fault manager (`fault_manager.c`) with hardware latching and locked-channel tracking:

DTC Code	Fault Identifier	Category	Priority	Exact Trigger Condition	System & Physical Hardware Reaction	Recovery / Reset Mechanism	Broadcasted CAN Frame
10..21	FAULT_CODE_OVERCURRENT_CH(0..11)	FAULT_CAT_HARDWARE	HIGH	Measured current on shunted channel i exceeds +130% of nominal rated limit (I_{max}) = [5A, 7.5A, 10A, 15A, 30A]). Canals 0..2, 4..8, 10..11 trip immediately. Inductive channels (Inverter CH9, Steering Wheel CH3) require ≥ 3 consecutive samples $> 130\%$ (inrush filter).	Immediately turns off the MOSFET by setting gate pin to HIGH (OFF), sets mosfets_status[i] = 0 , and engages channel latch via fault_manager_lock_channel(i) . All subsequent CAN override reactivation commands (ID 0x100) are rejected.	Requires power cycling the vehicle or issuing an authorized diagnostic reset command once the overcurrent condition has been resolved.	CAN IDs 0x001 / 0x00 (Status byte = CAN ID 0x501 (Byte = 1, Bytes 3-4 10+i, Byte 7 = Locked Chann Bitmask)
100	FAULT_CODE_VBAT_UNDERVOLTAGE	FAULT_CAT_HARDWARE	HIGH	Low-voltage battery voltage $V_{bat} < 5.0V$ continuously for $> 200\ ms$.	Cuts power to all 12 MOSFET channels simultaneously to protect the lithium auxiliary battery pack from catastrophic voltage collapse and cell degradation.	Automatic recovery when battery voltage rises back above 5.0V during pre-charge or vehicle power-up.	CAN IDs 0x001 / 0x00 (All 12 bytes = 0) CAN II 0x006 (Reports V_{ba} in mV) CAN ID 0x501 (Byte = 1, Bytes 3-4 100)
1	FAULT_CODE_CAN_ERROR_PASSIVE	FAULT_CAT_COMMUNICATION	LOW	TWAI peripheral alert TWAI_ALERT_ERR_PASS triggered by elevated error counters on the physical bus.	Internal diagnostic logging; non-intrusive warning transmitted to telemetry logging station.	Automatic when communication error rate drops.	CAN ID 0x50: (Byte 0 = 2, By 2 = 0, Bytes 3- = 1)
2	FAULT_CODE_CAN_BUS_OFF	FAULT_CAT_COMMUNICATION	HIGH	Controller enters BUS_OFF state or bus error count exceeds 50 (bus_error_count > 50).	Initiates automatic hardware recovery cycle via twai_initiate_recovery() . Retains all MOSFET safe states during bus reset.	Automatic once CAN controller successfully resynchronizes with the differential bus.	CAN ID 0x50: immediately upon recovery

3. CAN / TWAI Bus Network Architecture (500 kbps)

3.1. Transmitted CAN Frames

CAN ID	Message Name	DLC	Rate	Target Nodes	Payload Layout (Byte-by-Byte)	Resolution & Units
0x001	PDM_MOSFETS_1_8	8	10 Hz	Dashboard, MCU, Telemetry	Byte 0..7 : MOSFET 1 to 8 Status (1 = ON, 0 = OFF)	1\ Byte/Channel (Boolean)
0x002	PDM_MOSFETS_9_12	4	10 Hz	Dashboard, MCU, Telemetry	Byte 0..3 : MOSFET 9 to 12 Status (1 = ON, 0 = OFF)	1\ Byte/Channel (Boolean)

CAN ID	Message Name	DLC	Rate	Target Nodes	Payload Layout (Byte-by-Byte)	Resolution & Units
0x003	PDM_CURRENTS_0_3	8	10 Hz	Telemetry, Data Logger, MCU	Byte 0-1 : Current CH0 (Little-Endian uint16) Byte 2-3 : Current CH1 (Little-Endian uint16) Byte 4-5 : Current CH2 (Little-Endian uint16) Byte 6-7 : Current CH3 (Little-Endian uint16)	1\ LSB = 1\ mA Range: 0 \dots 65535\ mA
0x004	PDM_CURRENTS_4_7	8	10 Hz	Telemetry, Data Logger, MCU	Byte 0-1 : Current CH4 (Little-Endian uint16) Byte 2-3 : Current CH5 (Little-Endian uint16) Byte 4-5 : Current CH6 (Little-Endian uint16) Byte 6-7 : Current CH7 (Little-Endian uint16)	1\ LSB = 1\ mA
0x005	PDM_CURRENTS_8_11	8	10 Hz	Telemetry, Data Logger, MCU	Byte 0-1 : Current CH8 (Little-Endian uint16) Byte 2-3 : Current CH9 (Inverter) (LE uint16) Byte 4-5 : Current CH10 (Little-Endian uint16) Byte 6-7 : Current CH11 (Little-Endian uint16)	1\ LSB = 1\ mA
0x006	PDM_CURRENTS_HALL_VBAT	8	10 Hz	Telemetry, Data Logger, MCU	Byte 0-1 : Hall Shutdown (LE uint16 mA) Byte 2-3 : Hall Fans (LE uint16 mA) Byte 4-5 : Battery Voltage LV (LE uint16 mV) Byte 6 : Steering Wheel Warning (1 = L\vol > 2.5A) Byte 7 : Reserved (0)	Currents in mA, Voltage in mV (12600 = 12.6V)
0x501	PDM_DIAGNOSTIC_DTC	8	10 Hz / On-Fault	Safety Master, Dashboard, Telemetry	Byte 0 : High Fault Active (1 = Yes, 0 = No) Byte 1 : Category Byte 2 : Priority Byte 3-4 : Active DTC Code (Big-Endian uint16) Byte 5-6 : Fault Count Byte 7 : Locked Channels Bitmask (Bit i = CH i)	Bit 0: CH0, Bit 1: CH1, ..., Bit 11: CH11

3.2. Received CAN Frames

CAN ID	Sender Node	Key Payload Fields	PDM Action & Functional Behavior
0x021	MCU (Motor Control Unit)	Byte 6 : Vehicle State & R2D (4 = READY_TO_DRIVE).	Interlocks the OTA firmware update service, forbidding remote code flashing during vehicle operation.
0x100	Steering Wheel / Control	Byte 0 : Channel ID (0 \dots 11) Byte 1 : Command (1 = Enable, 0 = Disable).	Toggles MOSFET power for the specified channel, strictly rejecting activation if the channel is latched in fault mode.

4. Unit Test Validation Suite (Unity Test Framework)

All current monitoring equations, inrush filters, and battery protection debouncers are verified in PlatformIO native x86 simulation:

```
Processing * in native environment
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Building...
Testing...
test/test_main.cpp:95: test_vbat_conversion [PASSED]
test/test_main.cpp:96: test_protection_undervoltage_debounce [PASSED]
test/test_main.cpp:97: test_protection_overcurrent_fast_trip [PASSED]
test/test_main.cpp:98: test_protection_inrush_debouncing [PASSED]
test/test_main.cpp:99: test_fault_manager_channel_lock [PASSED]
----- 5 Tests 0 Failures 0 Ignored -----
OK
```

Detailed Test Case Specifications

- test_vbat_conversion :
 - Asserts battery voltage calculation using resistor divider ($R_2 = 4047.62\ \Omega$, $R_3 = 1100.0\ \Omega$, factor ≈ 4.6796). Confirms accurate 12.6V reading.
- test_protection_undervoltage_debounce :
 - Simulates a $< 5.0V$ voltage dip: verifies that transient drops lasting $< 200\ ms$ are ignored, while conditions exceeding $200\ ms$ trigger global channel shutdown and trip DTC 100 .
- test_protection_overcurrent_fast_trip :
 - Verifies instantaneous electronic fuse trip when channel current exceeds $1.30 \times I_{\{nom\}}$ on standard loads.
- test_protection_inrush_debouncing :
 - Verifies the 3-sample debouncing filter on capacitive/inductive channels (Inverter CH9 and Steering Wheel CH3): confirms that 1 or 2 high samples do not cause nuisance trips, and the channel trips reliably on the 3rd consecutive sample.
- test_fault_manager_channel_lock :
 - Validates that tripped channels are latched in the fault manager bitmask and reject subsequent unauthorized reactivation commands.

5. Industrial Standards & Safety Compliance Certification

- Zero Dynamic Memory:
 - 100% static allocation. Zero use of dynamic heap (malloc , free , new).
 - Static FreeRTOS queues and tasks pinned to Core 1 for telemetry serialization.
- Deterministic Hard Real-Time Timing:
 - Primary protection loop executes at strict 100 Hz (10.0 ms) via vTaskDelayUntil .
- MISRA-C:2012 Strict Conformance:
 - Fully qualified integer types, bounded loop counters, and explicit casts across all analog-to-digital conversions.
- Clean Compiler Verification:
 - Zero warnings under -Wall -Wextra -Werror compiler flags.